

Week 1 — Introduction to Business Forecasting

Business Forecasting · LIVE pilot (slides + exercises in one deck)

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Who am I?

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A decision you'll be able to make

You manage inventory for Walmart México. It's April. How many umbrellas should each store stock for June?

- Order too few → empty shelves, lost sales, unhappy customers.
- Order too many → cash tied up in stock that sits until next year.

The right number is a **forecast**: last year's June sales, the trend, the rainy-season pattern, a promotion you're planning — turned into a decision, with a sense of how uncertain it is.

By the end of this course, you'll build exactly this kind of model — on your own data.

Where this course is going

Every week is one step from *raw data* to a *forecast you can defend*:

C1 Framing decisions → **C2** describing data → **C3** quantifying uncertainty → **C4–C5** regression: what drives the outcome → **C6** time series & forecasting.

Keep this arc in mind: the statistics in the first half exists to make the **forecasts** in the second half trustworthy.

What Is This Class About?

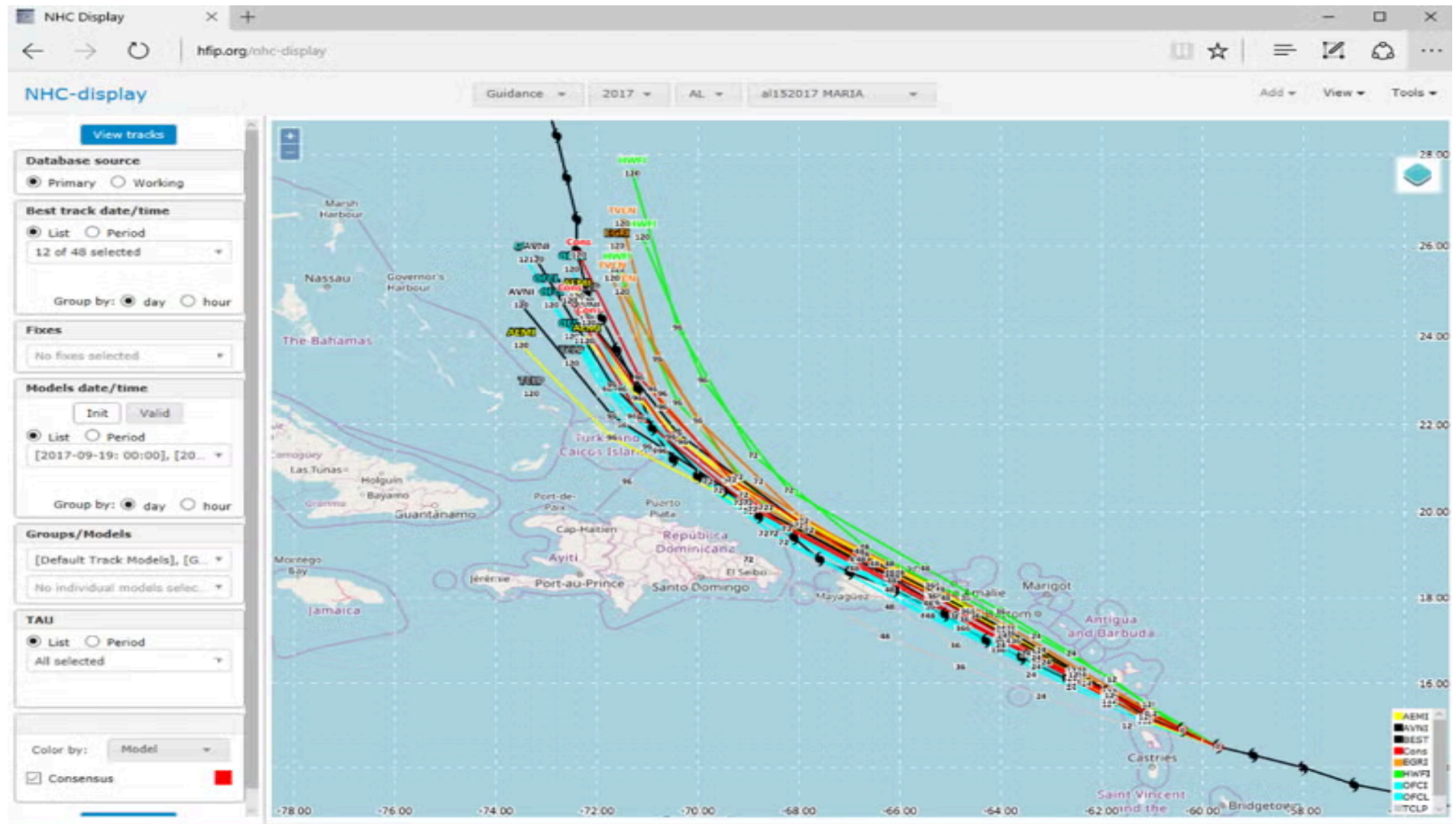
Forecasting is the science and art of making predictions about future events using historical data and relevant information.

In this class, you will learn how to turn data into actionable insights that drive better decisions in business and beyond.

Why is Forecasting Important?

1. Weather Forecasting

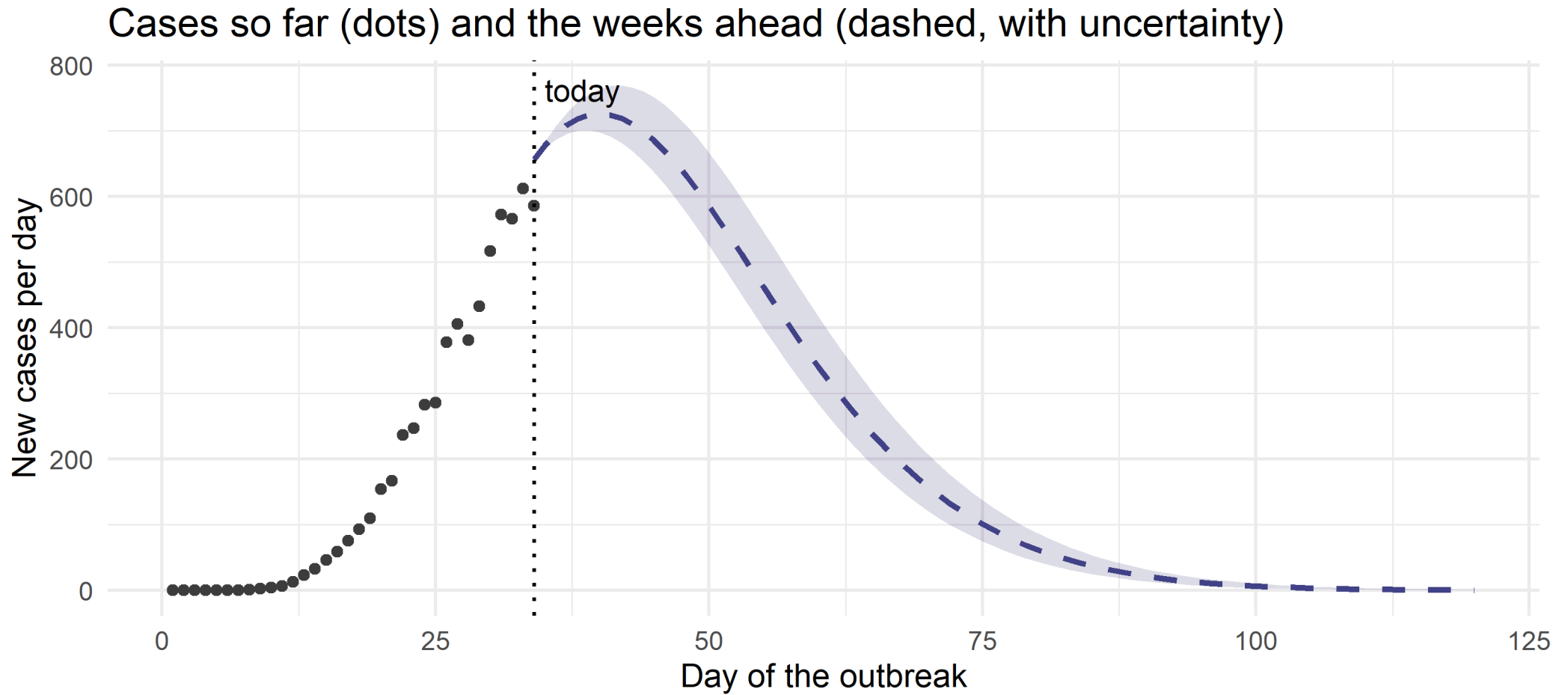
- Predicting hurricanes → timely reactions and preparations.



Epidemic Forecasting

2. Epidemic Forecasting

- Predicting disease outbreaks → effective public health responses.

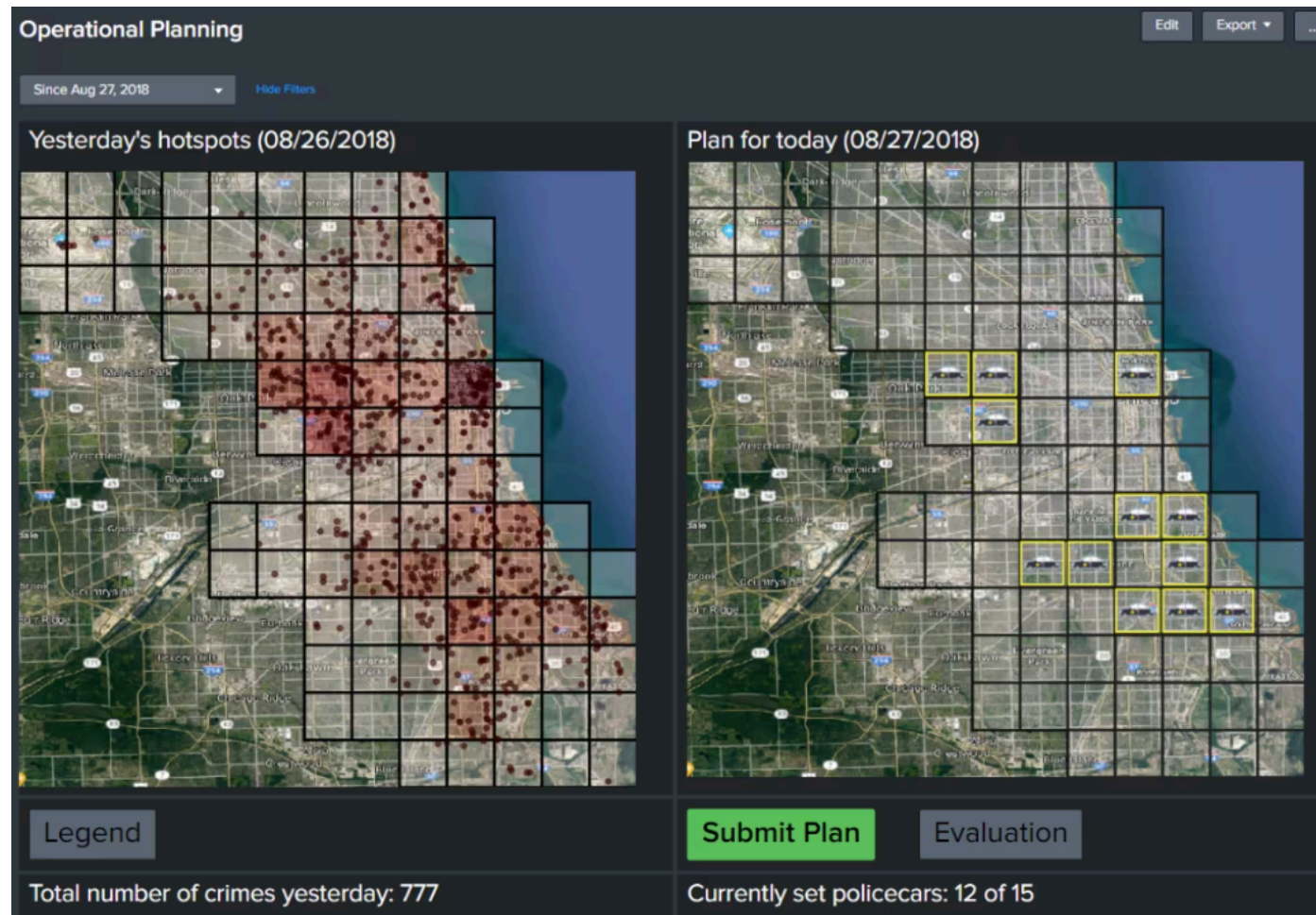


Where the peak lands, and how high, decides how many beds and staff you need.

Crime Forecasting

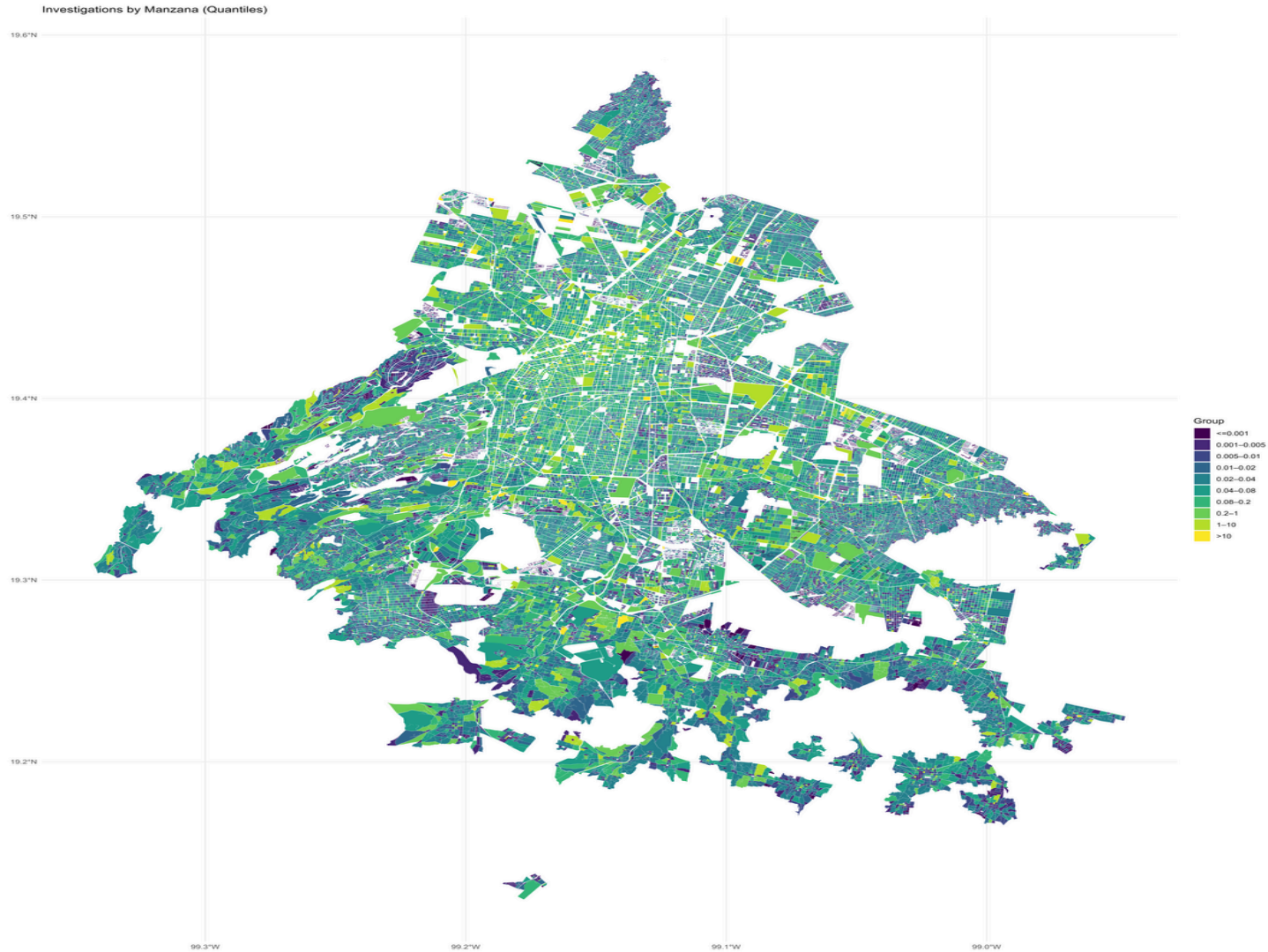
3. Crime Forecasting

- Identifying crime hot spots → more effective police patrols.
- Identifying individuals at risk → targeted interventions.



Predicting 911 calls in Mexico City

One of your tasks later in class: **Predict 911 calls in Mexico City**



Forecasting in Business

- With the increasing availability of data, firms rely on rigorous data-driven methods to make decisions.
- Many business decisions depend on anticipating future events.
- In this course, you'll learn how to use data to make informed predictions about the future.

Applications of Forecasting

1. **Anticipating Demand** Adjust inventory to minimize waste and meet customer needs effectively *Example: Meal kits, retail chains*
2. **Employee Churn** Predict which employees are likely to leave and develop strategies to retain talent *Example: HR analytics in consulting firms*
3. **Strategic Investment Decisions** Use market forecasts to guide investment timing and location *Example: Expansion of electric vehicle infrastructure*
4. **Improving Advertisement Effectiveness** Identify which components of an ad drive conversions *Example: A/B testing by Airbnb*
5. **Product Recommendations** Forecast customer preferences based on past behavior *Example: Amazon, Netflix recommendation systems*

Your project: pick a real problem, predict a real outcome

In this course you'll choose **your own topic and data**, build a regression model, and forecast a target I assign you — then defend it in a short oral exam. (Full details on Canvas; you pick topics in the first weeks.)

An example of what that looks like — a past cohort predicted daily public-transit ridership to help plan resources:

- How many vehicles are needed on a given day? How many drivers to schedule?
- Predict the **number of passengers on a specific future date**.
- Signals they used: school and holiday calendars, weather, city events, road repairs.

First the data they had 📱

The data behind the project

They assembled one row per day, over three school years:

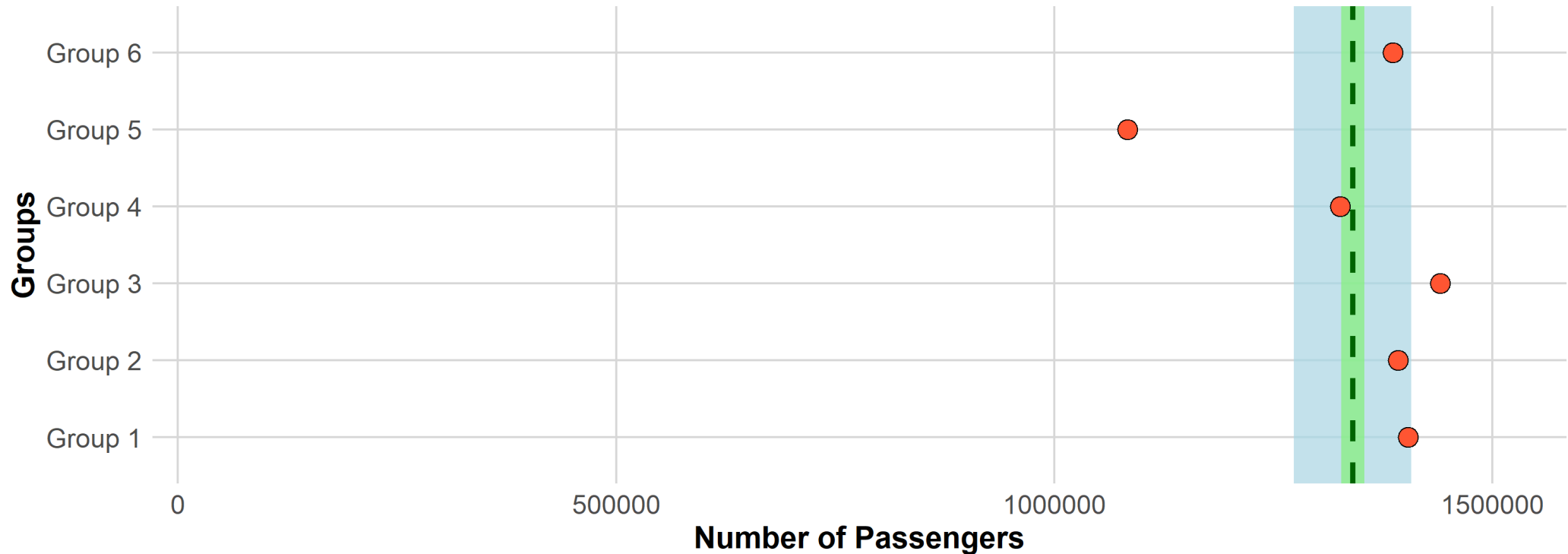
- the outcome — how many people rode the bus that day,
- the calendar — which date, which weekday, whether school was in session,
- the weather — rain, snow, temperature.

This is the same file you will load, summarise and plot yourself later in this class.

How close did the predictions get?

Predictions with 95% Confidence Intervals

Group-wise predictions with actual value as reference
Shaded areas: 2% (green) and 5% (blue) off actual value



Key Outcome

- Students built a **linear regression model** combining multiple data sources to generate forecasts.
- **Closest prediction was within 1% of the actual number of riders!**
- A reliable tool to improve resource allocation and decision-making.

Your turn

- Get in pairs
- Consider your past employment or your future employment
- Think about how forecasting could solve some problems in the context of industry you are considering
- (5 min)

What will you learn?

1. Getting Business Information from the Data

- Analyze data, evidence, and arguments to make reasoned judgments

2. Problem Solving and Forecasting

- Formulate, evaluate, and implement statistical models for business forecasting.
- Interpret the results and validate assumptions
- Key technical skill very valuable on the job market!

3. Decision Making and Communication

- Choose optimal options to achieve objectives.
- Communicate findings, conclusions, and recommendations effectively to business professionals

Organization

- **Lectures Schedule:**
 - Monday and Wednesday Morning
- **Textbooks**
 - See canvas course materials and syllabus
- **Organization**
 - see the website

Grading

- **70% Exams**
 - 3 exams (2 midterms + cumulative final), applied, in the computer lab
- **20% Group Project** (one project, all semester)
 - A semester-long project with an **individual oral defense** — you explain your own work
- **10% Quizzes & Datatons**
 - 3 short in-class Canvas quizzes + 2 live “Datatons” (forecasting challenges)
 - Each mirrors that week’s practice pack; **lowest score dropped**

Everything that counts is done in class or defended in person.

Pre-requisites

- Mathematics III or Linear Algebra I
- Statistics II or Statistical Inference

Language

- Class is in English
- But your English is not evaluated
- If you don't know a word, feel free to ask
- Great occasion to learn vocabulary useful for interviews

AI Tools

- Most of your grade — **the exams and the project defense** — is earned **without AI**. You won't have access to it there, so you need to genuinely **learn and understand** how to do these things on your own.
- You **can** use AI to **support your learning**: refining code, summarizing concepts, checking your understanding.
- Be honest with yourself, though — students who leaned heavily on AI **without challenging themselves** typically did **not** do well in this course.

Use AI to learn, not to replace thinking

- Great for review — try asking:
 - “Why can’t we say a 95% confidence interval contains the true value with 95% probability?”
 - “What’s the relationship between Type I and Type II errors?”
 - “What is a p-value and what is it not?”

AI and learning

Explain p- value to me as Taylor Swift



Alright, let's do this the Taylor Swift way!

🎵 Hey, it's me, Taylor Swift! Let's talk about p-values like we're writing a song together. 🎵

So, imagine you're writing a song about someone who you *think* might be into you. But you're not sure, so you're looking for signs—like how often they text you, or if they show up at your shows. Your null hypothesis (just a fancy way of saying your starting assumption) is, "They're just being nice; they don't really like me."

Now, let's say you collect a bunch of "data" (you know, how often they text, how long they stay at your shows, whether they're in the front row singing every lyric). You run the numbers, and out comes the p-value.

If the p-value is super low (like 0.01), that's like saying, "There's only a 1% chance that all these signs are just a coincidence." In Taylor Swift terms, this is the part in the song where you're like, "*Oh my God, it's fate! They totally like me!*" So you reject the null hypothesis and start planning your next big love song.

But if the p-value is high (like 0.5), that's more like saying, "There's a 50/50 chance they're just being nice and it's nothing special." In this case, you might be singing something more like "*We are never ever getting back together*" because there's just not enough evidence to believe they're into you.

In short, the p-value helps you figure out whether to go all-in on those feelings or if you should just shake it off!

Academic Integrity

- Integrity matters — for your own learning, for your peers, and for the course.
- **We will employ a set of tools to detect cheating. Each episode of cheating will be reported.**

Software: R

Performing Real Forecasts with Real Data

- **Widely Used** R is one of the most popular languages for data analysis, statistical modeling, and forecasting — especially in academia and industry.
- **Free and Open-Source** R is completely free to use and supported by a large global community.
- **Great Community Support** Thousands of tutorials, packages, and forums are available — it's easy to find help when you're stuck.
- **Powerful Data Visualization** R includes powerful tools (like `ggplot2`) to create clear, publication-quality visualizations that help you communicate your insights effectively.

Two places to run R — the same R

RStudio, on your laptop

- What you install this week
- Where you do swirl practice and the project
- **What you use in the exams**

These slides, in the browser

- Nothing to install, opens with the deck
- What we use to practice in class
- Your typing is saved in your browser

It is the **same language and the same code** — only the program around it differs. `mean(x)` is `mean(x)` in both.

Which means one thing for you

We practice **here** in class because it starts instantly and nobody gets stuck on installation.

You are examined **in RStudio**. So the R you install this week has to become familiar — that is what the weekly **swirl** lessons are for. Type the code in both places and the difference stops mattering.



Exercise 0 — The same code, right here

Prove it to yourself. Run these three lines below, then run the identical lines in RStudio later today — same answers, different window.

Exercise ↺ Start Over ⚠ Show Solution ▶ Run Code

```
1 2 + 2
2
3 x <- c(10, 20, 30, 40)
4 mean(x)
5
6 sqrt(144)
```

The bus ridership data — what is actually in it?

The transit dataset from the project example is loaded for you as `m`.

Nobody has told you what its columns are called or what values they take. Before using a dataset you always find that out yourself — with three commands.



Exercise 1 — Load and inspect data

Show the first rows of `m`, then the structure of every column, then a numerical summary of the whole data frame.

Type in the box and press Run — R runs in your browser.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # First rows:
2
3 # Structure of every column:
4
5 # Numerical summary of every column:
```

What `str()` just told you

Column	What it holds
<code>Bus_ridership</code>	passengers that day — the outcome to predict
<code>Date, Month, Year, Day_of_week</code>	the calendar
<code>School</code>	1 if school was in session, 0 otherwise
<code>PRCP, SNOW</code>	rain and snow that day
<code>TMAX, TMIN, TAVG</code>	highest, lowest and average temperature

`head()` shows values, `str()` shows types, `summary()` shows the range each column covers. Run all three on any new dataset.



Exercise 2 — Pick a column with \$

A whole data frame is rarely what you want — usually you want one column. Pull the `Bus_ridership` column out of `m` and show its first few values.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Pull one column out of the data frame and show its first values:
```

How big is this dataset — and what does `length()` count?

Write your guess down before running anything:

- Roughly how many rows does `m` have? How many columns?
- `nrow(m)` counts rows. What will `length(m)` give — rows, or columns?
- And `length(m$Bus_ridership)` — the same number, or a different one?



Exercise 2b — Check your guess

Run the commands and compare each result with what you wrote down.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

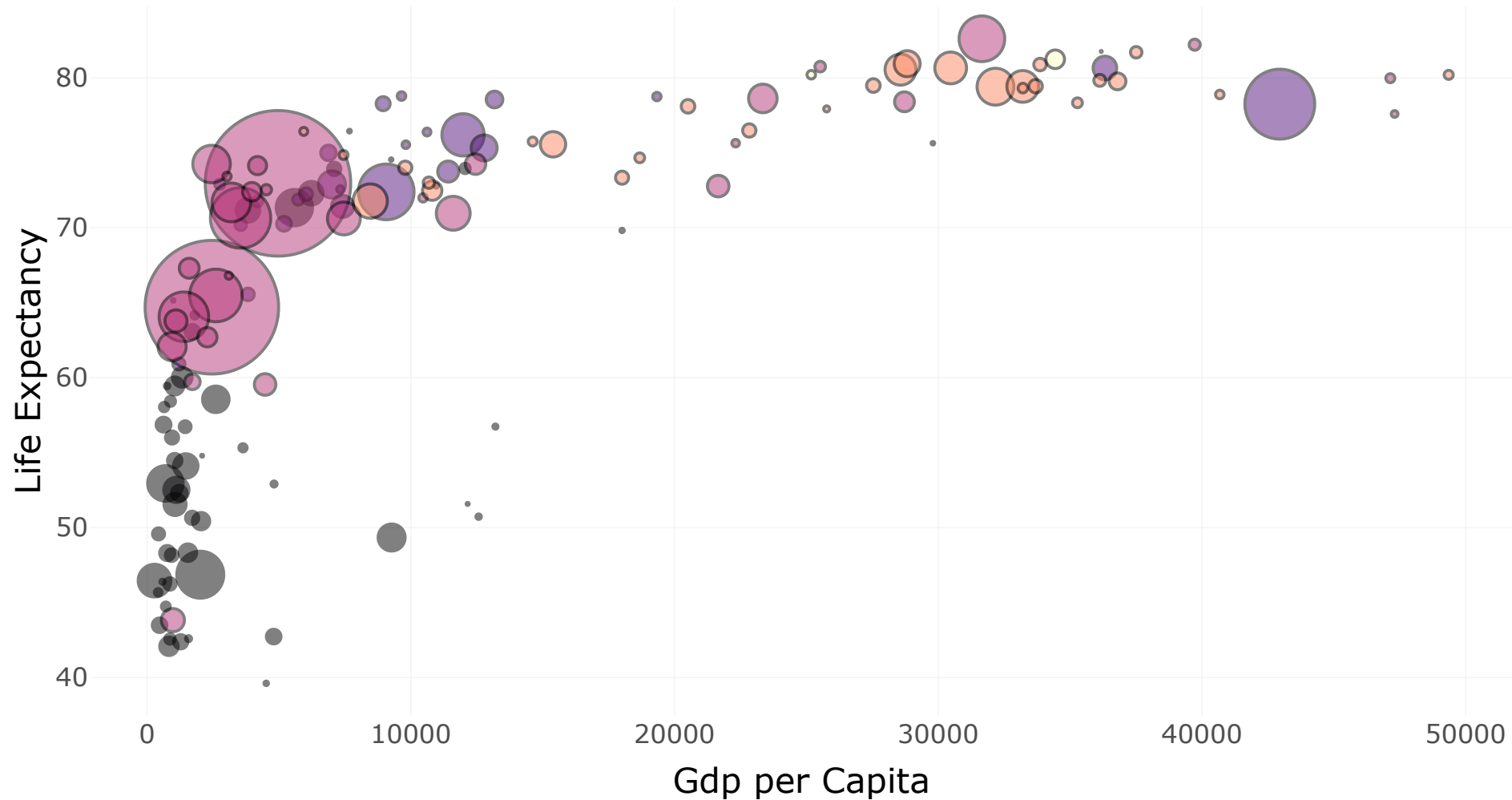
```
1 # Number of rows, and number of columns:
2
3 # length() applied to the whole data frame:
4
5 # length() applied to a single column:
```

What you just found

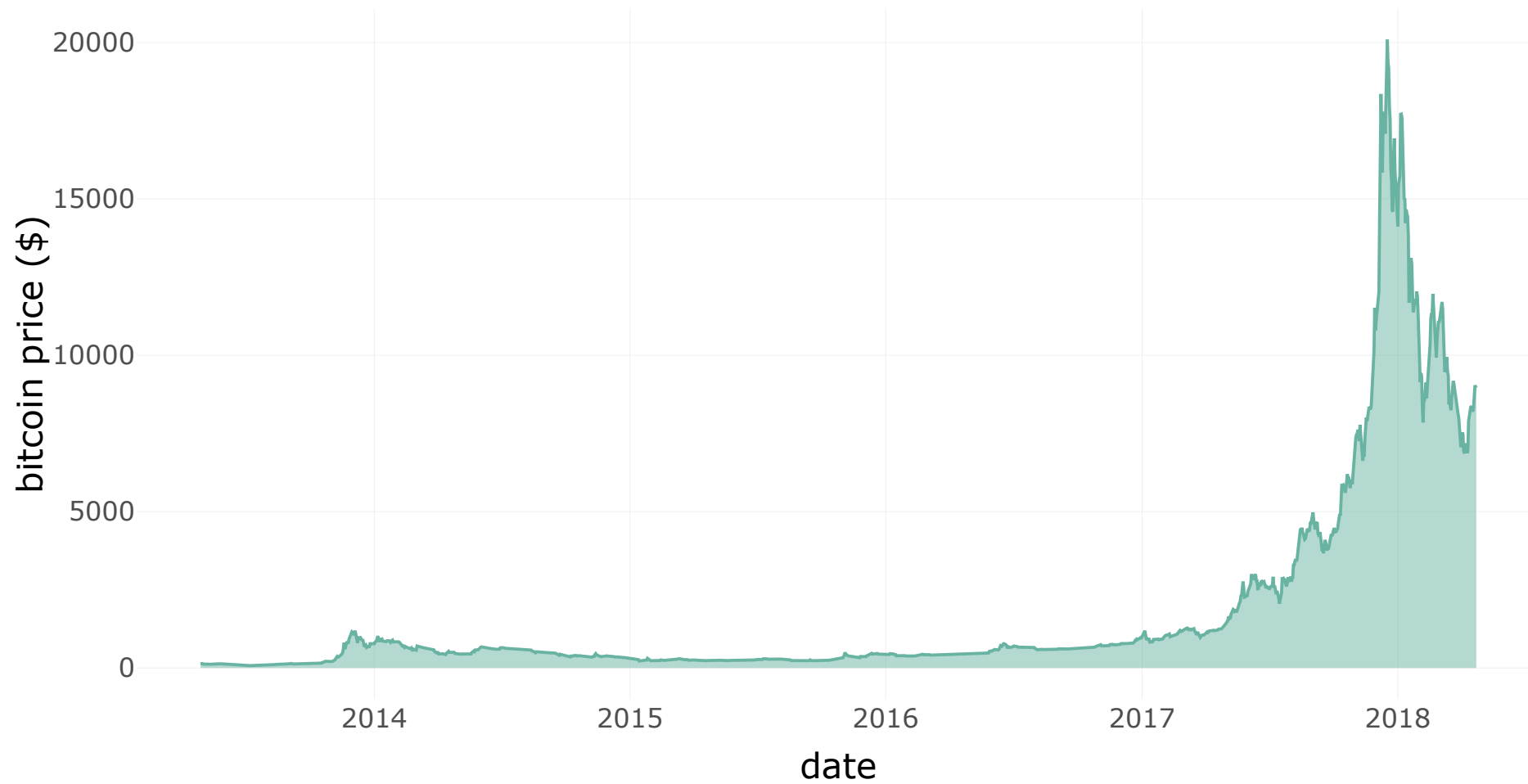
- A data frame is a **collection of columns**, so `length(m)` counts **columns** — it matches `ncol(m)`, not `nrow(m)`.
- A single column is a **vector**, so `length(m$Bus_ridership)` counts its elements: one per day, matching `nrow(m)`.
- `dim(m)` reports both, **rows first, columns second** — the same order as `m[rows, cols]`.

Same function, different object, different meaning. Always check what you are counting.

Life expectation vs GDP per Capita



Bitcoin price in time



Remarks

- Knowledge of theoretical statistics does not matter much if you can't apply it using modern tools
- We will use it for practical exercises with data
- You will use it for the final project
- You will use it for the applied exams
- We will learn some of it together in class
- TA lab sessions will further help with this
- Chatgpt is your friend



Exercise 3 — Basic vector arithmetic

Ridership is in passengers. Create a new vector expressing it in **thousands** of passengers by dividing `m$Bus_ridership` by 1000, and show the first few values.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Build the rescaled vector, then show its first values:
```



Exercise 4 — mean() and sum()

Compute the **average** daily ridership and the **total** ridership across all days.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Average daily ridership:
2
3 # Total ridership over all days:
```



Exercise 5 — min, max, and range

Find the **lowest** and **highest** daily ridership. Then get both at once with `range()`.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Lowest day, then highest day:
```

```
2
```

```
3 # Both at once:
```



Exercise 6 — Subsetting with [] and a condition

Keep only the days where ridership was **above one million** passengers. Subset the rows of `m` with a logical condition and look at the result.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Keep only the rows that satisfy the condition:
2
3 # Look at the result, and count how many days survived:
```

Introduction to Forecasting

Tools will often depend on the horizon and data availability

Forecasting Horizons

- **Very Short Horizon:**
 - High-Frequency Trading: Real-time price predictions for financial trading
- **Short Horizon:**
 - Public bikes: Predict the availability of bikes at bike station and adjust the number
- **Long Horizon:**
 - New obesity drug: Forecasting number of potential patients and their resources

Overview of Forecasting Techniques

1. Qualitative Forecasting

- Based on subjective judgment and expert opinions
- Suitable for unique situations or new markets
- Examples: Predicting economic impacts of oil price changes or political stability in a region

Methods of Qualitative Forecasting

The oracle of Delphi



Delphi Method

A structured process to reach consensus on complex, uncertain, long-term questions:

1. Select a group of experts.
2. Keep them **anonymous** — they don't talk to each other.
3. Ask them to answer a questionnaire.
4. Collect the initial responses.
5. Compile them into a summary.
6. Send the summary back; collect **refined** answers.
7. Repeat until consensus (or no further improvement).

Delphi Method — example

Determining AI threats.

- *Question:* what are the biggest risks from AI development?
- *Panel:* computer scientists, engineers, AI-company CEOs, ethicists.
- Send questionnaires on potential threats → compile → send back.
- Several rounds until the panel converges.
- *Output:* a ranked list of the most probable risks.

Brainstorming

A creative technique for generating ideas — free thinking, building on others.

- Form a group (**no need for experts**).
- State the problem clearly.
- Encourage ideas, no matter how crazy.
- Combine and build on each other's ideas.
- Document, then synthesize.

Example — employee engagement: a tech firm's HR gathers reps from HR, IT, and other teams to generate ideas for an app that boosts engagement, then implements the most promising ones.

Panel of Experts

Assemble knowledgeable people **in the same place, at the same time.**

- They meet, share insights, and discuss openly.
- Helps reach well-informed decisions.
- Often ends in a report with conclusions.

Example — environmental policy: a government agency convenes scientists, economists, conservationists, and policymakers to debate options and draft comprehensive environmental policy.

Focus Groups

Gather a **diverse** set of participants — not necessarily experts.

- Share perceptions, attitudes, and opinions.
- Produce qualitative data and consumer insight.

Example — a new TV show: before launch, a network gathers viewers from varied demographics to gauge appeal and fine-tune the show and its marketing.

The four methods — at a glance

One business use for each method:

- **Delphi** → ranking **AI threats** with an anonymous expert panel.
- **Brainstorming** → an HR team's ideas for an **employee-engagement** app.
- **Panel of Experts** → a convened panel drafting **environmental policy**.
- **Focus Groups** → viewers gauging a **new TV show** before launch.

Overview of Forecasting Techniques (cont.)

2. Quantitative Forecasting

- Uses historical data and numerical techniques
- Suitable when data is available and continuity assumptions hold true
 - *Continuity assumption*: past trends and relationships continue in the future
 - When it holds?
- Interest rates and investments
 - When it does not hold?
- Covid Cases & Deaths and Vaccines

Methods of Quantitative Forecasting

Time Series Forecasting

- **Time series data**: collection of data points for a single unit (one firm, one person, one country) ordered chronologically. Can be one or more variables.
- **Time series forecasting**: identifying patterns and trends in historical data to predict future values

Year	GDP per capita (USD)
2010	9400
2011	10342
2012	10376
2013	10866
2014	11076
2015	9753
2016	8875
2017	9434

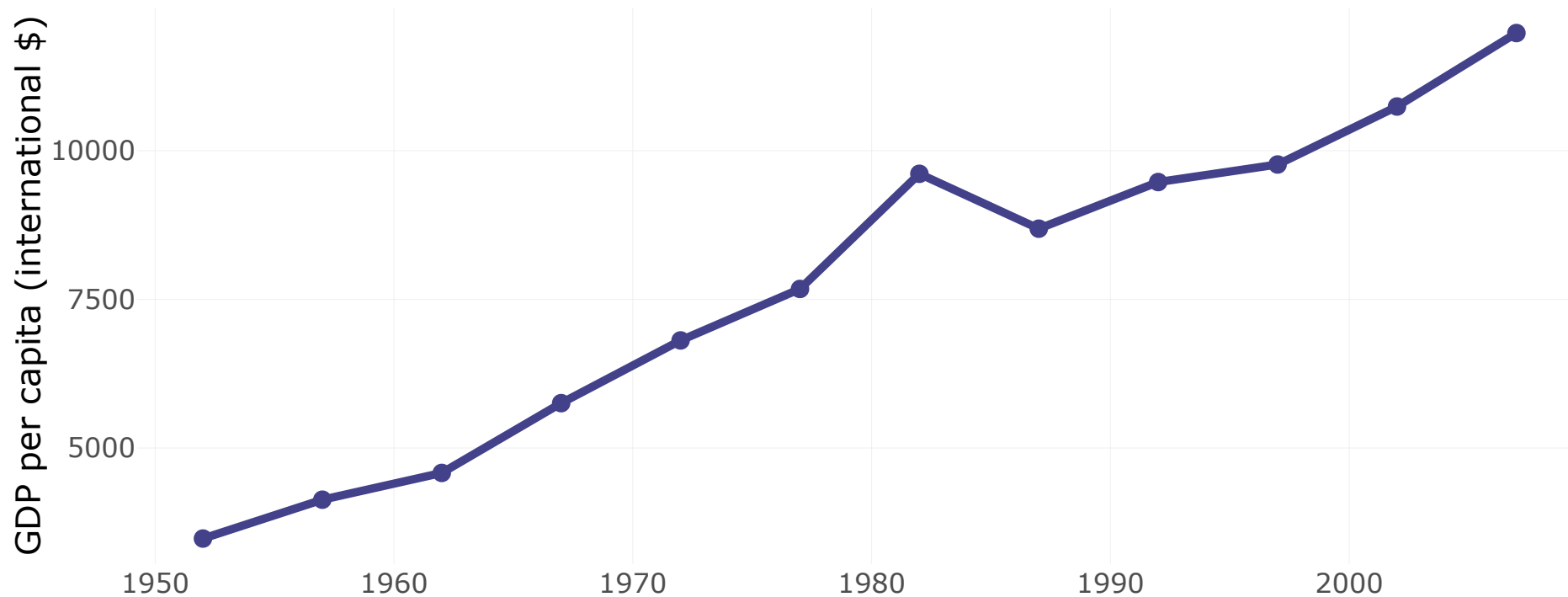
In simple terms:

- We don't care about what causes what
- We just hope that past values of the variable and its historical behavior can predict its future values

Example: Forecasting of GDP

- Time series forecasting can help predict a country's Gross Domestic Product.

Mexico: GDP per capita, 1952-2007



Hover any point. Given this history, what would you predict for the next five years?

Explanatory Models

- We have data on both the variable of interest and other variables related to it
- We consider how other variables impact the outcome of interest
- We use these relationships to make forecasts

Example: Sales at a new location

- Should we open a new Starbucks at ITAM?
- Using existing locations, analyze impact of:
 - Foot traffic
 - Neighborhood income
 - Competitors' stores
- Given these relationships, what would be sales at ITAM?

S_ID	Address	F_Traffic	N_Income	C_Stores	Sales
1	123 Main St	100	60000	2	5000
2	456 Oak Ave	150	75000	1	6000
3	789 Elm Rd	80	55000	3	4500
4	321 Maple Ln	120	80000	2	7000
5	543 Birch St	200	70000	1	8000
6	876 Pine Ave	90	60000	3	5500
7	987 Cedar Rd	180	90000	2	9000
8	654 Walnut Ln	110	85000	1	7500
9	234 Spruce St	140	75000	2	6500
10	ITAM	170	65000	1	NA

Steps of Forecasting

1. Problem Definition

- Clearly define the forecasting objective.

2. Gather Data

- Identify and collect relevant data.

3. Preliminary Explanatory Analysis

- Understand data characteristics and relationships.

4. Choosing and Fitting the Model

- Select and fit the appropriate forecasting model.

5. Evaluating the Model

- Assess the model's performance using historical data.

6. Communicate the results

- Visualize and interpret the outcomes and write a report or presentation

Remainder of the course

Quantitative Forecasting

- 0. Ungraded quiz
- 1. Review of Statistics
- 2. Simple linear regression
- 3. Multiple linear regression
- 4. Time Series

Is there a pattern over time?

You now have three school years of daily ridership, and a mean and a range for it. None of those numbers tell you whether ridership is flat, rising, falling, or repeating itself.

Only one thing will: put the days on an axis and look.



Exercise 7 — A base plot() over time

Plot `Bus_ridership` (y) against `Date` (x) with base `plot()`, and label the axes. Then describe out loud what you see.

Exercise

↺ Start Over

💡 Show Hint

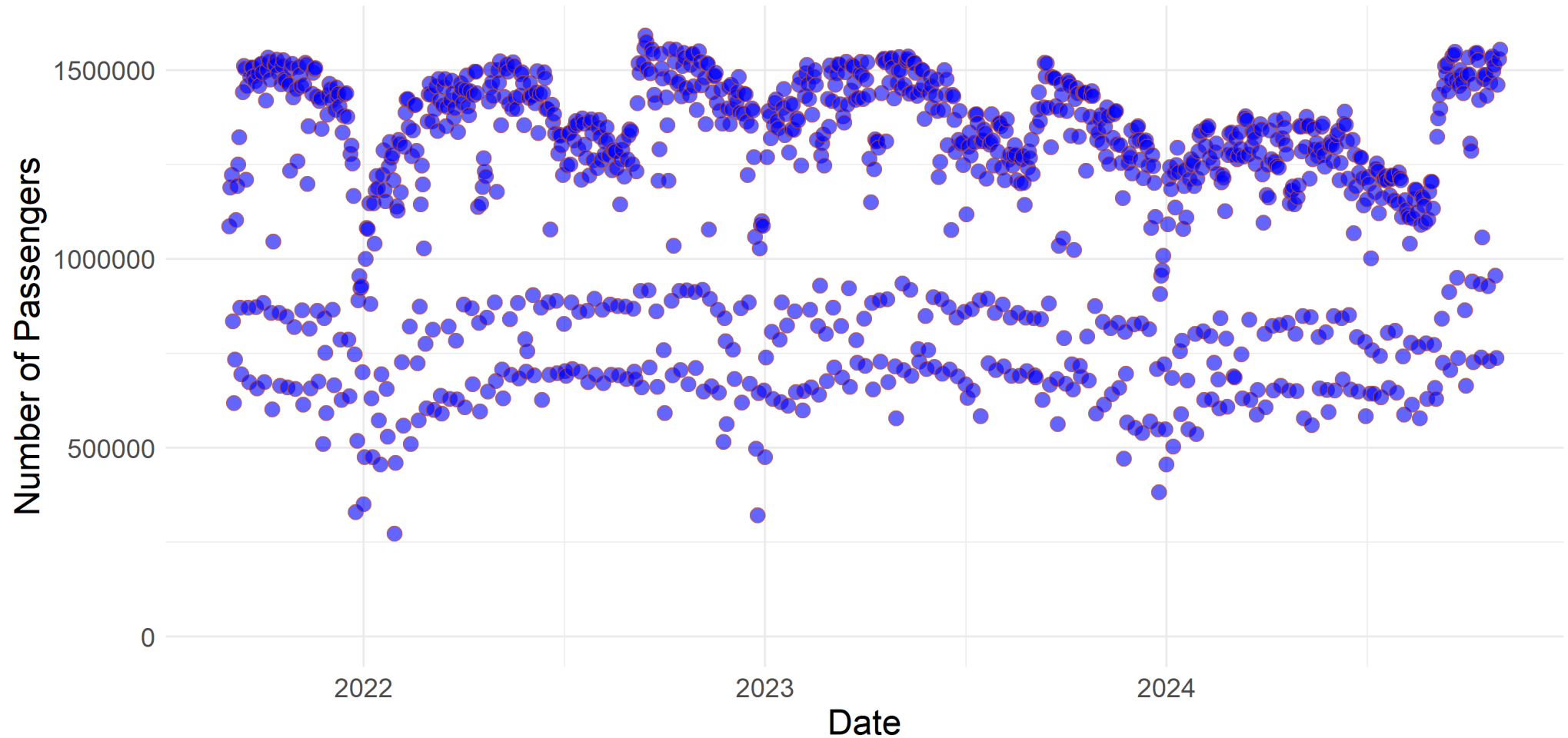
▶ Run Code

```
1 # Plot ridership against date, with labelled axes and a title:
```

Daily ridership — the polished version

Daily Bus Ridership in NYC

Trends in passenger numbers over time



Same data, same two variables as your plot — drawn with [ggplot2](#), which you meet later in the course. The structure you spotted is what makes this series forecastable.

Are all days alike?

The plot over time tells you **when** each value happened. It does not tell you **how often** each level occurs: are most days clustered near one level, or spread widely? Are there days far away from the rest?

That question needs a different picture.



Exercise 8 — A base hist()

Draw a histogram of `Bus_ridership`, labelled. Then say where most days sit and whether anything sits far from the crowd.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Histogram of daily ridership, with an axis label and a title:
```



Exercise 8b — Change one thing

The code below already works. Run it as it is. Then change `breaks` to `5`, run again; then to `100`, run again.

Answer in one sentence: what changed, and what did **not**?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 hist(m$Bus_ridership, breaks = 20,  
2     xlab = "Bus ridership", main = "Distribution of daily bus ridership")
```

Before next class — your Week 1 checklist

1. **Get R running.** Follow the **Week 1 Setup Guide** on Canvas: install R + RStudio, run the one package-install block, and open the first tutorial. Bring a working laptop — from Week 2 we use R live in class.
- 2. **Start the weekly swirl lesson.** This is where you practise in **RStudio** — the program you are examined in. The slides get you moving fast; swirl builds the habit.
- 3. **Bookmark the [R cheatsheet](#)** — you'll use it all semester (and in exams).
- 4. **Start thinking about a project topic** — a question you care about, with data you can get. Groups form in Week 2.

By the end of today you should be able to: say what forecasting is, give three business decisions it informs, and explain how this course is graded.

Questions?